WORKED SOLUTION

STEP II 2015, Question 11

Mechanics • Momentum and Energy

STEP II 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

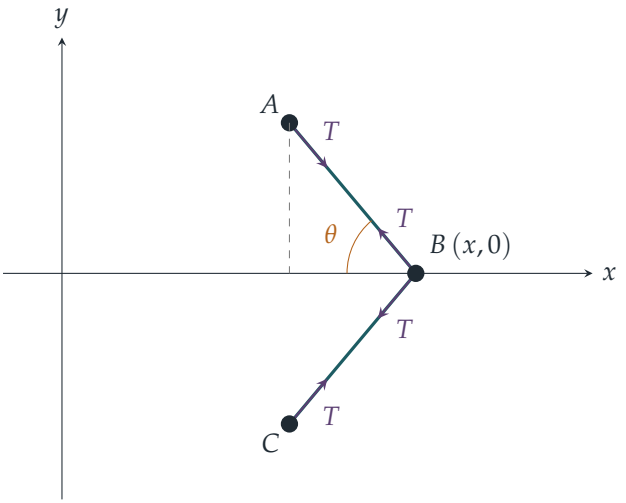
step.maths.org/questions/15-s2-q11

The key idea

Two coordinates, x and θ, describe the whole system, because the motion stays symmetric about the x-axis. The table is smooth and the string is light and inextensible, so two things are conserved: momentum in the x direction and kinetic energy. Those two conservation laws give the two equations in (ii) directly, and the second one, read carefully, says that $\dot{\theta}$ can never be zero. That single fact settles both (iii) and (iv).

Solution

We model A, B, C as particles and the string as light and inextensible. Gravity and the normal reaction are vertical, so the only horizontal forces are the tensions, which are internal to the system.



Part (i)

A is a distance a from B along a line making angle θ with the negative x direction, so

$$A = (x - a\cos\theta,\ a\sin\theta).$$

Differentiating with respect to t, using the chain rule on $\cos\theta$ and $\sin\theta$,

$$\dot{\mathbf{r}}_A = (\dot{x} + a\sin\theta\,\dot{\theta},\ a\cos\theta\,\dot{\theta}),$$

as required (and $\dot{x} = v$, the speed of B).

The set-up and the impulse are both symmetric under reflection in the x-axis, so the motion stays symmetric: $C = (x - a\cos\theta,\ -a\sin\theta)$ with velocity $(\dot{x} + a\sin\theta\,\dot{\theta},\ -a\cos\theta\,\dot{\theta})$.

Part (ii)

First, the starting values. At the instant of the impulse both strings lie along the y-axis, perpendicular to B's new velocity, so the string does not need to jerk A or C; they start at rest, while $\dot{x} = u$. The total x-momentum is mu and the kinetic energy is $\frac{1}{2}mu^2$.

Momentum. There is no external horizontal force, so the total x-momentum stays equal to mu:

$$m\dot{x} + 2m(\dot{x} + a\sin\theta\,\dot{\theta}) = mu \quad\Longrightarrow\quad 3\dot{x} + 2a\dot{\theta}\sin\theta = u. \tag{1}$$

(The y-momenta of A and C cancel, as they must.) The right-hand side here is u, the initial speed of B. The paper prints v, but v has already been used for $\dot{x}$, so this must be a misprint for u.

Energy. The string is light and inextensible, so the tension does no net work on the system, and the only other forces do no work either. Kinetic energy is conserved until A and C collide:

$$\dot{x}^2 + 2\Big[(\dot{x} + a\sin\theta\,\dot{\theta})^2 + a^2\cos^2\theta\,\dot{\theta}^2\Big] = u^2. \tag{2}$$

From (1), with $s = \sin\theta$ and $c = \cos\theta$,

$$\dot{x} = \tfrac{1}{3}(u - 2as\dot{\theta}), \qquad \dot{x} + as\dot{\theta} = \tfrac{1}{3}(u + as\dot{\theta}).$$

Substituting into (2), the cross terms $\mp 4uas\dot{\theta}$ cancel:

$$\tfrac{1}{9}(u - 2as\dot{\theta})^2 + \tfrac{2}{9}(u + as\dot{\theta})^2 + 2a^2c^2\dot{\theta}^2 = \tfrac{1}{9}(3u^2 + 6a^2s^2\dot{\theta}^2) + 2a^2c^2\dot{\theta}^2 = u^2.$$

So $a^2\dot{\theta}^2(\frac{2}{3}s^2 + 2c^2) = \frac{2}{3}u^2$, that is $a^2\dot{\theta}^2(s^2 + 3c^2) = u^2$. Since $s^2 + 3c^2 = 3 - 2s^2$,

$$\dot{\theta}^2 = \frac{u^2}{a^2(3 - 2\sin^2\theta)},$$

which is the second result (with the same reading of v as u).

Part (iii)

Because $3 - 2\sin^2\theta \leq 3$, part (ii) gives

$$\dot{\theta}^2 \geq \frac{u^2}{3a^2} > 0.$$

So $\dot{\theta}$ is never zero. Between collisions it is continuous, so it keeps the same sign and θ changes monotonically.

At the start A is at rest, so part (i) gives $\dot{x} + a\dot{\theta} = 0$, i.e. $\dot{\theta} = -u/a < 0$. So θ decreases steadily from $\frac{\pi}{2}$. A and C always have the same x-coordinate and y-coordinates $\pm a\sin\theta$, so they meet exactly when $\sin\theta = 0$. The first collision is at $\theta = 0$, behind B.

Ask yourself What does the collision do to $\dot{x}$ and $\dot{\theta}$?

At $\theta = 0$ both strings lie along the x-axis, and A and C have the same x-velocity and y-velocities $\pm a\dot{\theta}$. They approach along the y direction, so the impulse between them is along y. That is perpendicular to the strings, so no impulsive tension arises, and every x-velocity is unchanged. In an elastic collision between equal masses the velocity components along the line of centres are exchanged. So A's y-velocity changes from $a\dot{\theta}$ to $-a\dot{\theta}$: the collision simply reverses the sign of $\dot{\theta}$, and $\dot{x}$ stays the same.

Momentum and energy are the same after the collision as before, so (1) and (2), and hence both results of (ii), hold for the next stage of the motion too. Now $\dot{\theta} > 0$, and as before it cannot vanish. So θ increases steadily from 0, and the next value with $\sin\theta = 0$ is $\theta = \pi$. It gets there in a finite time, at most $\pi a\sqrt{3}/u$, because $\dot{\theta} \geq u/(a\sqrt{3})$. The second collision happens at

$$\theta = \pi,$$

with A and C meeting *in front of* B.

Part (iv)

Throughout the motion $\theta \in [0, \pi]$, so $\sin\theta \geq 0$. If $v = \dot{x} = 0$, then (1) gives $2a\dot{\theta}\sin\theta = u$, and squaring and using (ii),

$$4a^2\sin^2\theta \cdot \frac{u^2}{a^2(3 - 2\sin^2\theta)} = u^2 \iff 4\sin^2\theta = 3 - 2\sin^2\theta \iff \sin^2\theta = \tfrac{1}{2}.$$

So the possible values are $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$.

Ask yourself Squaring threw away a sign. Which sign do we actually need?

We squared $2a\dot{\theta}\sin\theta = u$, and since $u > 0$ we also need $\dot{\theta}\sin\theta > 0$, i.e. $\dot{\theta} > 0$. At $\sin\theta = \frac{1}{\sqrt{2}}$ we have $\dot{\theta}^2 = \frac{u^2}{2a^2}$, so $\dot{\theta} = \pm\frac{u}{a\sqrt{2}}$ and $2a\dot{\theta}\sin\theta = \pm u$. From (1):

- if $\dot{\theta} > 0$, then $\dot{x} = \frac{1}{3}(u - u) = 0$;
- if $\dot{\theta} < 0$, then $\dot{x} = \frac{1}{3}(u + u) = \frac{2}{3}u$.

So the answer is no. On the first swing (θ falling from $\frac{\pi}{2}$ to 0) θ passes $\frac{\pi}{4}$ with $v = \frac{2}{3}u$. Between the first and second collisions (θ rising from 0 to π) $v = 0$ at both $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$. After the second collision $\dot{\theta}$ is negative again, so on the swing back from π to 0 we have $v = \frac{2}{3}u$ at both values. The pattern then repeats: $v = 0$ at these angles only on the swings where θ is increasing.

Checks

- At $\theta = \frac{\pi}{2}$ initially, (ii) gives $\dot{\theta}^2 = u^2/a^2$, matching $\dot{\theta} = -u/a$ found directly from A being at rest.
- From (1), $\dot{x} = \frac{u}{3}\left(1 - \frac{2a\dot{\theta}\sin\theta}{u}\right)$. At $\theta = 0$ this is $\frac{u}{3}$: all three particles move together at $\frac{u}{3}$, which is the centre-of-mass velocity, as it should be when the strings are straight along the x-axis. On the rising swing, at $\theta = \frac{\pi}{2}$, $\dot{x} = -\frac{u}{3}$, so B moves backwards for a while, which explains why v passes through 0 twice.
- A computer simulation of the three particles joined by very stiff springs, with $a = u = 1$, reproduces both equations of (ii). It shows the first collision at $\theta = 0$ and the second at $\theta = \pi$, and v changing sign only near $\theta \approx 0.785$ and 2.356 on the rising swing, where $\dot{\theta} \approx 0.707$. On the first swing it gives $v \approx 0.67$ at $\theta = \frac{\pi}{4}$.

Common slips

- Taking the right-hand side of (1) to be $\dot{x}$: it is the constant u fixed by the impulse.
- Assuming the collision sends θ back up towards $\frac{\pi}{2}$ and the second collision is again at $\theta = 0$. Reversing $\dot{\theta}$ makes θ increase *through* $\frac{\pi}{2}$, and nothing can stop it before π, because $\dot{\theta}^2$ is bounded away from 0.
- Forgetting that squaring loses the sign of $\dot{\theta}$ in part (iv).

Beyond STEP

Choosing x and θ as coordinates and building the kinetic energy from them is the starting point of Lagrangian mechanics, which you meet in the first year. There, conservation of x-momentum appears automatically because x does not occur in the energy (it is an "ignorable" coordinate). The energy equation then reduces the problem to a single first-order equation for θ, just as here. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers momentum and energy methods for systems of particles very thoroughly.

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